

TEJESWARA REDDY DUDYALA

9110727080 | dtejaswarreddy10@gmail.com | [linkedin.com/in/tejeswara-reddy-dudyala-74784821b](https://www.linkedin.com/in/tejeswara-reddy-dudyala-74784821b)

Portfolio : <https://dtejaswarreddy10.github.io/tejeswarareddydudyala>

Experience - 2 Years 9 Months

Design Engineer

Nov 2025 - Present

ACL Digital [Client - AMD]

Hyderabad

- Developed RTL-based verification test cases in Verilog/VHDL to validate new Vivado Synthesis features and language enhancements.
- Created verification plans and feature validation strategies for synthesis engine qualification.
- Debugged synthesis issues and analyzed RTL-to-netlist behavior across multiple language constructs.
- Managed and maintained regression suites for feature validation and quality tracking.
- Developed AI-assisted automation solutions for regression maintenance, test case mapping, report generation, and email notifications.
- Built AI agents for automated test plan review and validation workflow acceleration.

RTL Design Engineer

Jan 2024 - July 2025

Krisemi Design Technologies[1 Year 7 Months]

Bengaluru

- Designed FSM-based RTL modules for DVB-S2/DVB-S2X communication IP.
- Upgraded Frame Length Enforcer, LDPC and PL processing modules from DVB-S2 to DVB-S2X standards.
- Performed module-level and subsystem-level verification using Vivado simulation.
- Participated in architecture understanding, RTL development, integration, and protocol compliance verification.
- Contributed to complete system-level verification of DVB-S2/DVB-S2X IP.

Design Verification Trainee

June 2023 - November 2023

Moschip Institute of Silicon Systems[6 Months]

Hyderabad

- Designing of 16 bit ALU and 8 bit Customized up-down counter using Verilog and Verification using SV.
- Verification of APB based 10/100 ETHERNET and AMBA-3 APB Protocol by building the UVM Verification architecture from scratch to the functional coverage and ABV.

Skills

HDL : Verilog, VHDL
HDVL : System Verilog
EDA Tools : Questasim, VCS, Vivado, Onespin, Riviera, Xcelium
Protocols Worked on : APB3, 10/100 APB Based ETHERNET, DVB_S2 and DVB_S2X, UART
Protocols Known : AXI4, SPI, DMA, FIFO
Scripting : Python, TCL with the help of LLM's
Concepts : FSM Based Designing , Lint , CDC, Synthesis, Clock Gating, ASIC FLOW
Version Control : Github, Perforce
AI & Automation Tools : Claude Code (CLI), Claude Opus, OpenAI Codex, DEVStudio, Agentic Flows.

Projects

Mixed - Precision Sparsity-Aware Systolic Array [Currently Working]

June 2026

- Designed a parameterized output-stationary $N \times N$ INT8 systolic array in Verilog RTL for efficient computation.
- Optimized design to achieve ~506 MHz Fmax on Kintex-7 (xc7k70t-1), delivering a $2.7\times$ boost over baseline.
- Managed full RTL-to-routed flow in Vivado, utilizing out-of-context synthesis for timing and placement optimization.
- Developed a bit-accurate golden model in Python to validate hardware results through self-checking tests..

In-Memory Computing (IMC) SRAM Engine [1 Month]

March 2026

- Designed a 256×32 dual-port SRAM-based IMC engine in Verilog to accelerate edge AI/ML inference by performing bitwise computations directly within memory, addressing the von Neumann memory-wall bottleneck.
- Verified the design with self-checking testbenches (directed + randomized tests) and synthesized for resource utilization and throughput analysis.

MIPS / RISC-V Processor Design (Single-Cycle & Pipelined)- [3 Months]

Sept 2025 - Nov 2025

- Designed a 32-bit MIPS/RISC-V processor supporting R, I and J-type instruction formats.
- Implemented both single-cycle and 5-stage pipelined architectures (IF, ID, EX, MEM, WB).
- Added hazard detection, data forwarding logic to ensure correct pipelined execution.
- Developed ALU, Register File, Control Unit, PC, Branch logic, Memory Interface modules.
- Verified processor execution using waveform analysis in ModelSim/QuestaSim.

Systolic Array-Based Matrix Multiplication Accelerator [1 Month]

Oct2025

- Designed a parameterized 2×2 systolic array for matrix multiplication in Verilog RTL
- Built a reusable Multiply-Accumulate (MAC) processing element with systolic data forwarding, running accumulation, and async reset
- Instantiated and interconnected four MAC PEs to compute matrix products in parallel using spatial dataflow.
- A self-driving Verilog testbench written with clock/reset generation and corner-case stimulus for verification.
- Applied scalable, parameterized design (data/accumulator widths) to extend the architecture beyond 2×2.

UART TX/RX Design (With & Without FIFO Support) - [1 Month]

July 2025 - Aug 2025

- Designed Tx and Rx modules with configurable baud rate, data bits, parity, and stop bits.
- Implemented two architectures: direct TX/RX datapath and FIFO-based UART for improved throughput and reduced CPU load.
- Developed sync FIFO using circular buffer logic to enable buffered serial communication.
- Performed loopback testing to validate end-to-end TX→RX data integrity using QuestaSim.

Development Of DVB - S2 and S2X IP -Inhouse Project - [1Y 7M]

Jan 2024 - July 2025

- Designed Mode Adaptation Module in the DVB-S2 IP (Opted FSM based design).
- Designed CRC8 calculator for 8 bit input.
- Understanding the DVB-S2 and DVB_S2X Protocols.
- Upgraded the 3 modules(Frame_length_enforcer,LDPC,PL) RTL from DVB-S2 version to DVB-S2X version
- Module by Module Verification of DVB-S2 & DVB-S2X by Simulation on Vivado.
- Total DVB-S2 and DVB-S2X System Verification

APB based ETHERNET Protocol - Internal Project - [3 Months]

Oct 2023 - Dec 2023

- Understanding the Ethernet architecture and functionality.
- Preparation of the Verification Plan, Test plan, Coverage plan and Assertion plan.
- Involving the IP level verification using UVM environment for Ethernet IP Core.
- Implementation of the test scenarios to verify the functionality of Ethernet IP Core.
- Code coverage and Functional coverage analysis and assertion based formal verification.

AMBA 3 APB Protocol - Internal Project - [1 Month]

Sept 2023 - Oct 2023

- Understanding the APB protocol specification.
- Preparation of the Verification Plan, Test plan, Coverage plan and Assertion plan.
- Protocol verification with a single master and single slave in both SV and UVM.
- Implementation of the Functional coverage model and Assertions.

UP-DOWN COUNTER - Internal Project - [1 Month]

Aug 2023 - Sept 2023

- Understanding the specification.Implemented the up down Design.
- Verified the Design in SV Environment.Prepared the test, verification and coverage plan.
- Implemented Coverage and Assertions to check the functionality of DUT.

ALU - Internal Project - [1 Month]

July 2023 - Aug 2023

- Understanding the specification..Implemented the ALU design.
- Verified with a basic test bench.Prepared the design plan.

Bachelor's Academic Projects:

- **Control and Analysis of BLDC Motor using Different Control Techniques.**
- **Switching Control of Home Appliances by IR Remote using Microcontroller.**

Education

Birla Institute of Science and Technology,Pilani

Online

Master of Technology - Work Integrated Learning Program

Jan 2025 - Dec 2026

Coursework: VLSI Design & Microelectronics

Ongoing**R V R & J C College of Engineering**

Guntur,AP

Bachelor of Technology - Full Time

Aug 2019 - April 2023

Coursework: Electrical and Electronics Engineering

CGPA : 9.13